

ANIRUDH MOHOLKAR

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EDUCATION

University of Illinois at Urbana-Champaign | Dean's List | James Scholar | Fiddler Innovation Award

Expected 2027

Computer Engineering — Grainger College of Engineering

GPA: 3.79/4.0

Relevant Courses: Computer Systems & Programming, Data Structures & Algorithms, Distributed Systems, Artificial Intelligence, Applied Machine Learning, Reinforcement Learning

EXPERIENCE

Zebra Technologies — Software Engineer Intern, AI & Cloud Platform

May 2026 – Present

- **Agentic AI & BOM Integration:** Architected and deployed an enterprise RAG conversational agent within the Zebra Data Lake (ZDL) via BigQuery/Databricks; indexed proprietary BOM blueprints so analysts can query **alternate suppliers and transition schemes** in natural language during supplier blackouts, **eliminating manual cross-referencing** and safeguarding downstream fulfillment for **Amazon, Target, Walmart** and other tier-1 clients.
- **Autonomous Infrastructure Optimization:** Diagnosed cluster bottlenecks by analyzing production metadata, identifying a **234% concurrent load peak** on P3 nodes and 27K monthly unviewed report refreshes ("Zombie Reports"); engineered an autonomous agent deployed on **GCP Cloud Run** utilizing **Databricks Unity Catalog, PostgreSQL, and the MS Graph API** to negotiate and reschedule report timing directly with owners via Microsoft Teams.
- **Algorithmic Load Balancing:** Designed a custom **greedy optimization algorithm** leveraging **quadratic and exponential weightage schemes** to mathematically rank and re-distribute optimal data refresh windows, reducing projected cluster peak loads **under 55% capacity** and saving **\$11K/month**.
- **LLM Evaluation & Observability:** Built a proprietary prompt evaluation harness with 150+ test cases, optimizing agent **accuracy from 71% to 94%** and reducing hallucinations by 60%.

Zebra Technologies — Software Engineer Intern, Cloud & Computing

May 2025 – Aug. 2025

- **Distributed Infrastructure Automation:** Engineered an asynchronous Python and PowerShell data pipeline to extract and process system metadata, optimizing runtime from 2.5 hours to under 2 minutes: **98.5% latency reduction** across 10,000+ endpoints.
- **Backend Integration & Security Engineering:** Implemented a scalable data synchronization layer aggregating ServiceNow REST APIs and internal CMDB databases, automating vulnerability detection pipelines to isolate **3,386 security vulnerabilities** and eliminating 40 manual audit hours per cycle. Automation work **directly commended by Zebra's Chief Information Officer (CIO)**.

PROJECTS/EXTRACURRICULARS

Design for America – UIUC Chapter | Director of Operations, Executive Board

May 2024 – Present

- **Technical Product Steering:** Managed SDLC and architecture across 7 teams of 56 total members, directly leading 2 teams. Translated community requirements into high-performance MVPs, shipping 2 to live partners for 100+ users.
- **Full-Stack Engineering Blueprint:** Engineered an open-source data workshop utilizing a **Java 17 Spring Boot REST API**, JPA, and a JS client to standardize decoupled architecture; implemented strict separation of concerns via **DTOs/Mappers** and a centralized **@ControllerAdvice** global exception framework.

Distributed File System Development (HyDFS) | Go, gRPC, HTTP

Aug. 2025 – Dec. 2025

- **Collaborative System Design:** Co-developed a fault-tolerant distributed file system via hybrid RPC/HTTP architecture and gossip-based membership protocol, achieving **99%+ node failure detection accuracy**.
- **Consistency & Fault Tolerance:** Maintained a replication factor of 3 via background merges and automated re-replication; secured metadata integrity and seamless data recovery during concurrent node crashes and rejoins.
- **Stream Processing:** Built a distributed engine with leader-driven scheduling and bidirectional RPC logging, benchmarked against Spark Streaming using **500K-user workloads** with **<100ms latency operations**.

Operating Systems Kernel Development | C, RISC-V

Jan. 2025 – May 2025

- **Virtual Memory & Scheduling:** Built a custom RISC-V operating system kernel in C, developing a virtual memory manager with dynamic page allocation and a preemptive process scheduler.
- **Concurrency & Low Latency Performance:** Optimized page allocation and address translation, **reducing memory overhead by 28%** under concurrent workloads while diagnosing and resolving low-level race conditions.

Chicagoland Help Forum | Founder | React, Node.js, GitHub Actions CI/CD

Dec. 2020 – May 2024

- **Community Impact:** Founded and scaled a community platform to 300+ users, **raising \$5,400+** for under-served students with 100% donor transparency. Sustained over 3 years while enrolled full-time.
- **Technical Implementation:** Designed and maintained a responsive full-stack web platform generating **13,000+ page views** in 2024; shipped continuous feature updates via GitHub Actions CI/CD pipeline.

TECHNICAL SKILLS

AI & Agentic Systems: Agentic Workflows (ReAct/Reflection), LangChain, RAG Architecture, Vector Databases (Pinecone, Milvus), Prompt Evaluation (Evals), LLM Observability

Distributed Systems & Infrastructure: Fault-Tolerant Architecture, Microservices, Consensus Protocols (Raft/Paxos), Docker, Kubernetes, AWS (EC2, Lambda, Bedrock), CI/CD Pipelines

Low-Level & Systems: Linux Kernel Development, Multithreading/Concurrency, Memory Management, SystemVerilog, RISC-V/SLC-3 Architecture, Socket Programming

Languages & Tools: Python, C, C++, Java, JavaScript, SQL, Git, gRPC, REST APIs, Jira